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Sobo, a Recently Amplified Satellite Repeat of Potato, and Its Implications for the Origin of Tandemly Repeated Sequences

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ABSTRACT

Highly repetitive satellite DNA sequences are main components of heterochromatin in higher eukaryotic genomes. It is well known that satellite repeats can expand and contract dramatically, which may result in significant genome size variation among genetically related species. The origin of satellite repeats, however, is elusive. Here we report a satellite repeat, Sobo, from a diploid potato species, *Solanum bulbocastanum*. The Sobo repeat is mapped to a single location in the pericentromeric region of chromosome 7. This single Sobo locus spans ~360 kb of a 4.7-kb monomer. Sequence analysis revealed that the major part of the Sobo monomer shares significant sequence similarity with the long terminal repeats (LTRs) of a retrotransposon. The Sobo repeat was not detected in other Solanum species and is absent in some *S. bulbocastanum* accessions. Sobo monomers are highly homogenized and share >99% sequence identity. These results suggest that the Sobo repeat is a recently emerged satellite and possibly originated by a sudden amplification of a genomic region including the LTR of a retrotransposon and its flanking genomic sequences.

TANDEMLY repetitive DNA elements are frequently referred to as satellite DNAs because they were first isolated from satellite bands upon centrifugation in density gradients. Satellite DNAs are major constituents of large complex eukaryotic genomes. As a classical example, the HS satellite of the kangaroo rat *Dipodomys ordii* contains TTAGGG as its principal constituent and comprises 30% of the genomic DNA (Hatch and Mazrimas 1974; Fry and Salser 1977). Satellite sequences can diverge rapidly and their evolutionary patterns often do not follow any obvious phylogenetic order. The HS satellite of *D. ordii* in a close relative, *D. deserti*, is barely detectable (Mazrimas and Hatch 1972; Hatch and Mazrimas 1974). Over 160 families of satellite repeats have been described in higher plants (Macas *et al.* 2002). Satellite sequences reported in plants often show erratic distributions and great differences in their abundance among closely related species (Schmidt and Heslop-Harrison 1993; Kubis *et al.* 1997; Macas *et al.* 2000; Frello *et al.* 2004). Some satellite elements, however, are nearly universal within taxa. For example, the 2D8 satellite repeat discovered in potato is present in almost all species throughout the genus Solanum (Stupar *et al.* 2002).

It has been well established that satellite repeats are often associated with heterochromatic features and located mainly in the centromeric and telomeric regions. There is increasing evidence that many satellite repeats are structural or functional components of eukaryotic chromosomes. Satellite DNAs are often the major DNA components of the functional centromeres of complex eukaryotic species (Henikoff *et al.* 2001; Jiang *et al.* 2003). Nevertheless, the origin of satellite repeats has been elusive. Several hypotheses were proposed to explain the birth of satellite DNAs. A tandem repeat may be derived from nonrepetitious sequences by repeated and random unequal crossing overs (Smith 1976) or may be generated by both replication slippage and unequal crossing over and subsequently expanded by amplification mechanisms, which may create a new satellite (Walsh 1987). Salser *et al.* (1976) proposed a "library" hypothesis to explain the evolution of satellite repeats among genetically related species. According to this hypothesis, related species share a common library of satellite sequences. Specific members within the library may be amplified in some species and are present at low levels in other species (Salser *et al.* 1976; Fry and Salser 1977). Therefore, the appearance of most "new" satellites may not occur *de novo*, but due to amplification of one of the satellites present at a low level in the "library" (Salser *et al.* 1976).

Potato provides an excellent model to study the evolution of repetitive DNA elements. The asexual propagation of the potato species may minimize meiotic mechanisms that would remove newly emerged repeats (Stupar *et al.* 2002). *Solanum bulbocastanum* ($2n = 2x = 24$) is a diploid tuber-forming wild species distributed throughout central and southern Mexico to Guatemala (Spooner *et*

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al. 2004). This species has been regarded as self-incompatible on the basis of extensive pollination studies of different *S. bulbocastanum* accessions (Graham *et al.* 1959). Because of the paucity of seed balls on plants under natural conditions, asexual production through tubers is probably the main mode of propagation of this species.

Here we report the isolation and characterization of a tandem repeat specific to *S. bulbocastanum*. This repeat, named Sobo, was investigated with both molecular and cytogenetic approaches. Our results suggest that the Sobo repeat has emerged recently and is possibly derived from a DNA amplification event. The implications of the Sobo repeat for the origin of satellite DNAs are discussed.

MATERIALS AND METHODS

Plant materials: *S. bulbocastanum* clone PT29 (PI 243510) was used in cytological and DNA analyses. Additional accessions or clones of the three subspecies of *S. bulbocastanum* (ssp. *bulbocastanum*, ssp. *dolichophyllum*, and ssp. *partitum*) were provided by James Bradeen (University of Minnesota) or obtained from the Potato Introduction Station (Sturgeon Bay, WI) (Table 1). The haploid clone USW1 ($2n = 2x = 24$), derived from *S. tuberosum* cv. Katahdin, was used in Southern blot analysis. DNA of other Solanum species used in this study, including *S. etuberosum*, *S. ochranthum*, *S. morelliforme*, *S. bulbocastanum*, *S. cardiophyllum*, *S. lesteri*, *S. capsicibaccatum*, *S. chacoense*, *S. boliviense*, *S. infundibuliforme*, *S. agrimonifoluim*, *S. albornozii*, *S. verrucosum*, *S. multidissectum*, *S. oplocense*, *S. curtilobum*, *S. fendleri*, *S. iopetalum*, and tomato (*Lycopersicon esculentum*), were kindly provided by David Spooner, U.S. Department of Agriculture-Agricultural Research Service and Department of Horticulture, University of Wisconsin (Madison, WI).

Cloning, sequencing, and sequence analysis: A bacterial artificial chromosome (BAC) clone 28G12 was isolated from a *S. bulbocastanum* library (Song *et al.* 2000). BAC DNA was isolated using a standard alkaline lysis method and was digested with *Hin*dIII. The digested DNA fragments were resolved on an agarose gel. The two *Hin*dIII fragments, 3.8 kb and 866 bp, respectively, were cloned into pUC18 vector. One 3.8-kb clone and one 866-bp clone were completely sequenced using the GPS-1 Genome Priming System (New England Biolabs, Beverly, MA) following the manufacturer's instructions. Ends of additional clones from both fragments were also sequenced. Sequence analyses were performed using the Lasergene 99 software package (DNAStar, Madison, WI), dotter (Sonnhammer and Durbin 1995), and Staden package (Staden 1996). Sequence similarity searches were performed using BLAST and FASTA (Pearson and Lipman 1988; Altschul *et al.* 1997), and detection of conserved protein domains was done using RPS-Blast (Marchler-Bauer *et al.* 2003).

Fluorescence *in situ* hybridization: The fluorescence *in situ* hybridization (FISH) procedures were described previously (Jiang *et al.* 1995; Jackson *et al.* 1998). Probe DNA was labeled with biotin-dUTP or digoxigenin-dUTP in standard nick translation reactions and was detected with avidin-FITC and antidigoxigenin-rhodamine. Images were digitally captured with a CCD camera attached to an Olympus microscope and a Macintosh computer. The length of the fiber-FISH signals were digitally measured using IPLab software and converted to kilobases using a conversion ratio of 3.21 kb/μm (Cheng *et al.* 2002). Plasmid pTa71, which contains the coding sequences for the 18S·26S ribosomal RNA genes of wheat (Gerlach and Bedbrook 1979), was used as a rDNA probe.

Southern blot hybridization: Genomic DNAs were isolated from young leaves of greenhouse-grown plants. Restriction enzyme digestions of the DNA samples were performed and fractionated by agarose gel electrophoresis. DNAs were transferred to Hybond-H^+ membrane (Amersham Biosciences, Piscataway, NJ). Prehybridization and hybridization experiments were performed at 65°. The membranes were hybridized with ^{32}P-labeled DNA probe. After an overnight hybridization, membranes were sequentially washed with 2× SSC plus 0.1% SDS, then with 1× SSC plus 0.1% SDS, and finally with 0.5× SSC plus 0.1% SDS for 20 min each. The membranes were exposed to X-ray films and developed.

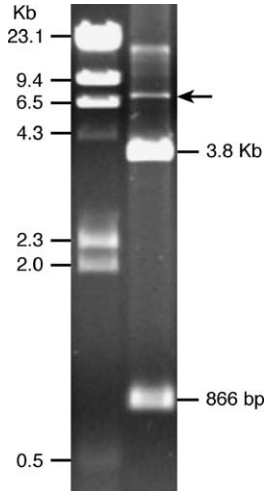


Figure 1.—Agarose gel electrophoresis of BAC 28G12 following *Hin*dIII digestion (right lane). BAC 28G12 produces two bands with sizes of 3.8 kb and 866 bp, respectively. Arrow points to the BAC vector.

RESULTS

Isolation and cytological characterization of the Sobo repeat from *S. bulbocastanum*: In an effort to characterize the major repetitive DNA families in the potato genome, we screened a BAC library of *S. bulbocastanum* using the genomic DNA of *S. bulbocastanum* as a probe. A BAC clone, 28G12, showed strong hybridization. BAC 28G12 contains a 150-kb insert based on pulsed-field gel electrophoresis (PFGE) (data not shown). *Hin*dIII digestion of 28G12 DNA produced only two bands, 866 bp and 3.8 kb, respectively, in addition to the vector band (7.4 kb) (Figure 1). This result suggests that BAC 28G12 contains a tandemly repeated element, which we named the "Sobo" (***So****lanum bul****bo****castanum*) repeat.

FISH analysis using BAC 28G12 as a probe revealed that this BAC hybridizes to only 1 of the 24 *S. bulbocastanum* (PI 243510; clone PT29) chromosomes (Figure 2, A–C). This hemizygous Sobo locus was mapped to the pericentromeric region of chromosome 7, which was identified by co-FISH mapping using the chromosome 7-specific BAC clone 38O2 (Dong *et al.* 2000) (Figure 2, A–C). Fiber-FISH was used to estimate the DNA size of the Sobo locus (Figure 2D). To better estimate the variation caused by the variable stretching degree of the

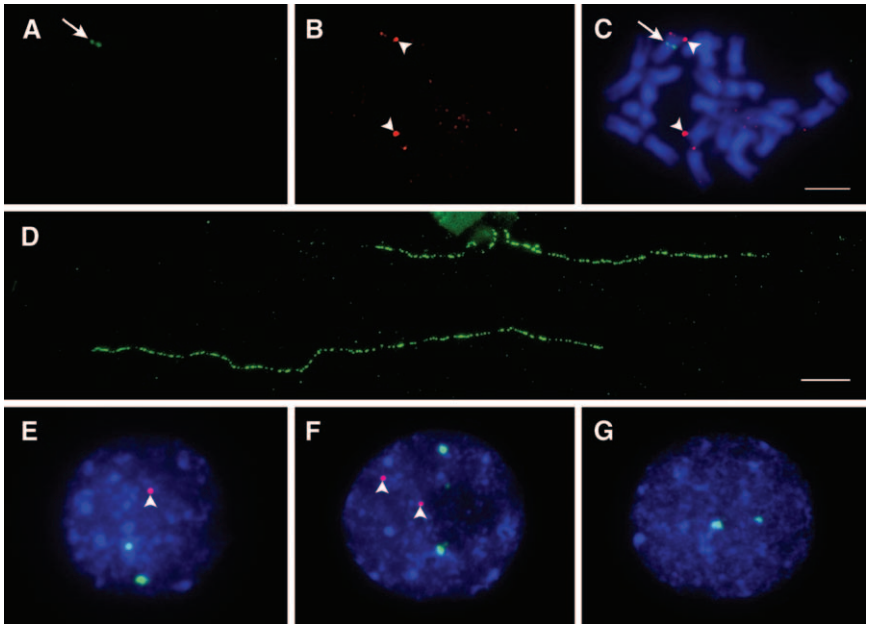


FIGURE 2.—Cytogenetic mapping of the Sobo locus in *S. bulbocastanum*. (A) A single FISH signal derived from BAC 28G12 (arrow) was observed in the *S. bulbocastanum* accession PI 243510 (clone PT29). (B) FISH signals (arrowheads) derived from the chromosome 7-specific BAC clone 38O2. (C) FISH signals are merged with the somatic metaphase chromosomes of PT29. The Sobo locus (arrow) is located in the centromere-proximal region of chromosome 7. Bar, 5 μm. (D) A microscopic field containing two fiber-FISH signals derived from the Sobo repeat in *S. bulbocastanum* accession PI 243510 (PT29). (E–G) Representative interphase nuclei containing 1, 2, or 0 FISH sites (red signals, arrowheads) from *S. bulbocastanum* accessions PI 275185, PI 498223, and PI 347757, respectively. The 18S·26S ribosomal RNA genes (green signals) were used as a reference probe. Bar, 10 μm.

genomic DNA fibers, we included the known BAC clone 220C03 in the same cytological preparations as an internal control. BAC 220C03 contains ~100 kb of single-copy sequences based on PFGE analysis (BRADEEN *et al.* 2003). The average length of 16 fiber-FISH signals derived from BAC 220C03 was 34.6 μm (data not shown), which converts into 111 kb of DNA using the conversion rate of 3.21 kb/μm (CHENG *et al.* 2002). This estimation is close to 100 kb estimated by PFGE, indicating that the 3.21 kb/μm conversion ratio is relatively accurate. The average length of 28 fiber-FISH signals derived from the Sobo repeat was 112 ± 38 μm. Using the 3.21 kb/μm conversion rate, we estimate that the Sobo repeat spans 360 ± 122 kb.

The Sobo repeat shares sequence similarity with the long terminal repeat of the *Sore1* retrotransposon: The two *Hin*dIII fragments, 866 and 3829 bp, respectively, were subcloned into the pUC18 vector. One plasmid clone derived from each of these two fragments was fully sequenced. Restriction mapping revealed that these two fragments are from a single monomeric unit and that the 866-bp fragment represents the first part of the monomer (data not shown). Thus, these two fragments are combined as a single 4689-bp Sobo element (AY849929 in GenBank) by adding the 866- and 3829-bp elements and subtracting a 6-bp *Hin*dIII cloning site in the junction.

Dot-plot analysis and sequence similarity searches of the Sobo monomer revealed a complex structure, including inverted and direct subrepeats within the monomer and regions of homology to several different genomic sequences (Figure 3). The region between nucleotides 368–837 is homologous to a segment of a BAC clone from *S. bulbocastanum* chromosome 8 (AY303171 in GenBank, nucleotides 145173–145650, FASTA expectation value $4.4e^{-30}$); this sequence is of unknown origin and does not show similarity to any other entry in the GenBank. The region between nucleotides 3390 and 4100 containing tandem subrepeats is highly similar (expectation value $1.3e^{-34}$) to the satellite repeat Sb4A of *S. brevidens* (PREISZNER *et al.* 1994).

The majority of the Sobo monomer (nucleotides 860–3380 and 4300–4688; Figure 3, A and C) displays similarity to a long terminal repeat (LTR) of a retrotransposon identified in a genomic sequence of *S. demissum* (AC 149290, position 123425–138040). This element, designated as *Sore1* (*So*lanum *re*trotransposon *1*), is 14,616 bp long, contains LTRs of 2699 and 2857 bp, respectively, and is flanked by a 5-bp target site duplication (TSD). On the basis of the sequence divergence of its LTRs (85.6% sequence similarity) and the presence of a mutation within the TSD and numerous mutations that interrupt open reading frames coding for retroelement proteins (data not shown), this *Sore1* element may be an ancient copy. In addition, there is an insertion of a part of an inverted LTR sequence within the coding region of the element (Figure 3D). However, detection of conserved protein domains using RPS-BLAST with intact segments of the element-coding regions allowed us to identify putative gag, reverse transcriptase, and integrase domains (expectation values of $4e^{-10}$, $6e^{-10}$, and $1e^{-11}$, respectively). The order of these domains indicated that this element belongs to the Ty3/*gypsy* group of retrotransposons.

We sequenced the ends of four additional plasmid clones derived from the 866-bp fragment and detected no sequence polymorphism. Similarly, we sequenced the ends of seven additional clones derived from the 3829-bp fragment. No polymorphism was detected among the ~300 bp that spans the 3′ sequence ends. The 5′ sequence ends of the seven clones, however, can be di-

vided into two subgroups. Sequences within each subgroup are identical. The sequences between two subgroups contain eight single nucleotide polymorphism sites scattered within a 580-bp region. Collectively, the sequencing data show that the monomers of the Sobo repeats share >99% sequence identity.

The Sobo element is present only in *S. bulbocastanum*: To survey the distribution of the Sobo repeat, we conducted Southern blot analysis of DNA from tomato and 19 Solanum species using the entire Sobo monomer sequence as a probe. None of these species, except *S. bulbocastanum*, hybridized to the Sobo repeat even under low hybridization stringency (Figure 4). We also conducted a separate Southern blot experiment including only *S. tuberosum, S. bulbocastanum*, and *S. demissum* and the Sobo element was not detected in *S. demissum* (data not shown). We further analyzed nine different accessions of three subspecies of *S. bulbocastanum* (*S. bulbocastanum* ssp. *bulbocastanum*, ssp. *dolichophyllum*, and ssp. *partitum*). The Sobo repeat was detected in six of the seven *S. bulbocastanum* ssp. *bulbocastanum* accessions (Figure 5).

We hypothesized that the absence of the Sobo repeat in certain *S. bulbocastanum* accessions may be due to segregation of the hemizygous Sobo locus in natural populations. We conducted interphase FISH analysis of seven additional clonally maintained accessions (Table 1, lines 1–7). No signals were observed in four accessions (Table 1, Figure 2G). The other three accessions contain one or two hybridization sites (Table 1, Figure 2, E and F). We also obtained seeds of six accessions (Table 1, lines 8–13), which were also included in Southern blot analysis (Figure 5). FISH analysis using a population of 10 seeds from each accession was conducted. FISH results confirmed that two of the six accessions (PI 347757 and PI 275200) do not contain the Sobo repeat (Table 1). All of the seeds from two accessions (PI 275169 and PI 498223) contain two signals, presumably representing a homozygous Sobo locus (Table 1). Furthermore, two of the six accessions (PI 275185 and PI 275192) displayed variation in the number of Sobo FISH signals from 0 to 2. This indicates that the parents of these seeds likely had a hemizygous Sobo constitution and the number of Sobo foci were therefore segregating in these populations. Collectively, the FISH results in these *S. bulbocastanum* accessions confirmed that the Sobo repeat is present as a single locus and is segregating in some populations.

The Sobo sequences are heavily methylated: Most satellite repeats in higher eukaryotic genomes are heavily methylated. We used restriction combinations of *Eco*RII/*Bst*NI and *Hpa*II/*Msp*I to test if the Sobo sequences are methylated. Methylation-sensitive *Eco*RII and methylation-insensitive *Bst*NI cleave the CC(A/T)GG site, depending on the methylation status of the interior

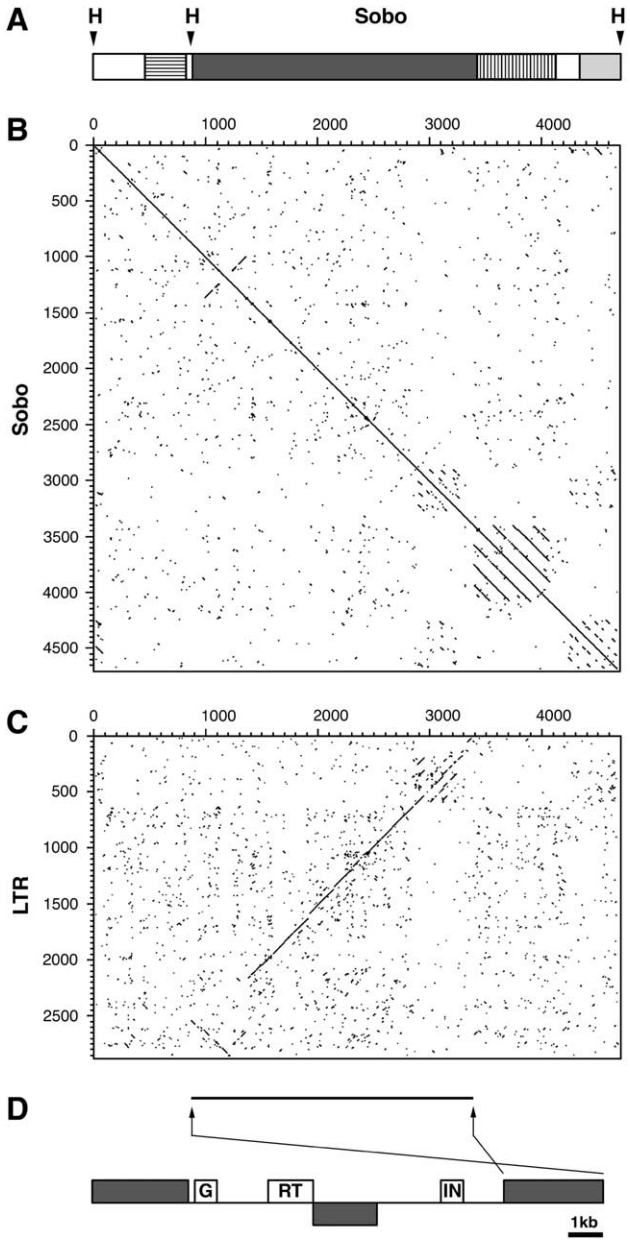


FIGURE 3.—Structure of the Sobo monomer sequence. (A) Schematic of the Sobo monomer showing positions of two *Hin*dIII sites (H) and regions of homology to the *S. bulbocastanum* sequence AY303171 (horizontal lines), the satellite repeat Sb4A of *S. brevidens* (vertical lines), and the LTR of retrotransposon *Sore1* of *S. demissum* (dark shading). The 3′-end of the Sobo monomer includes imperfect tandem repeats with scattered homology to the structurally similar region within the LTR of *Sore1* (light shading). (B) The dot-plot comparison of the Sobo sequence with itself and (C) with the right LTR of the *Sore1* element. Sequence similarity >67% (at least 40 identical nucleotides within a window of 60) is represented with a dot or diagonal line. (D) A scheme of the *Sore1* element in AC149290 showing positions of the LTR sequences (shading; the insertion of a partial LTR sequence within the element is in reverse orientation) and putative gag (G), reverse transcriptase (RT), and integrase (IN) coding domains. Horizontal line and arrows under the dot-plot show position and orientation of the LTR-like sequence within the Sobo element.

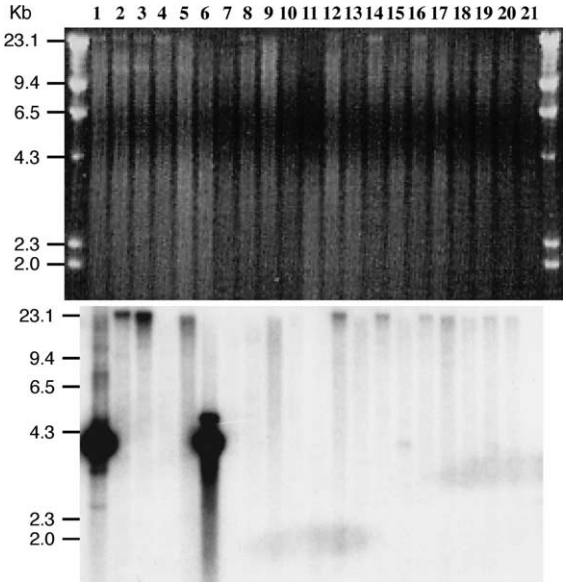


FIGURE 4.—Distribution of the Sobo repeat in Solanum and related species. *Hin*dIII-digested DNAs from *S. bulbocastanum* PI 243510 (PT29) (lane 1), *S. tuberosum* (USW1) (lane 2), *S. etuberosum* PI 558303 (lane 3), *S. ochranthum* PI 230519 (lane 4), *S. morelliforme* PI 275223 (lane 5), *S. bulbocastanum* PI 275198 (lane 6), *S. cardiophyllum* PI 347759 (lane 7), *S. lesteri* PI 442694 (lane 8), *S. capsicibaccatum* PI 205560 (lane 9), *S. chacoense* PI 414153 (lane 10), *S. boliviense* PI 545963 (lane 11), *S. infundibuliforme* PI 498246 (lane 12), *S. agrimonifoluim* PI 243352 (lane 13), *S. albornozii* PI 498206 (lane 14), *S. verrucosum* PI 275260 (lane 15), *S. multidissectum* PI 210043 (lane 16), *S. oplocense* PI 435079 (lane 17), *S. curtilobum* PI 225649 (lane 18), *S. fendleri* PI 498004 (lane19), *S. iopetalum* PI 275182 (lane 20), and tomato (*L. esculentum*) PI 1334 (lane 21) were hybridized with the Sobo repeat. Only two *S. bulbocastanum* accessions (lanes 1 and 6) show hybridizations.

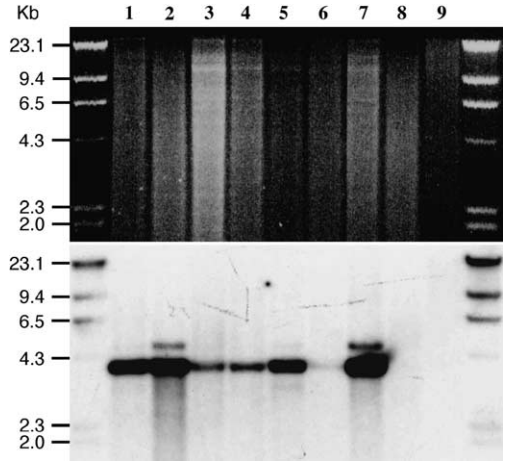


FIGURE 5.—Southern analysis of a set of nine *S. bulbocastanum* accessions. *Hin*dIII-digested DNAs from *S. bulbocastanum* ssp. *bulbocastanum* PI 243510 (PT29) (lane 1), PI 275198 (lane 2), PI 275185 (lane 3), PI 275192 (lane 4), PI 275196 (lane 5), PI 347757 (lane 6), PI 498223 (lane 7), *S. bulbocastanum* ssp. *dolichophyllum* PI 595473 (lane 8), and *S. bulbocastanum* ssp. *partitum* PI 275200 (lane 9) were probed with the Sobo repeat.

cytosine. Similarly, *Hpa*II and *Msp*I recognize the CCGG sequence, but neither can cut when the 5′ C is methylated. Only *Msp*I, not *Hpa*II, can cleave if the internal C is methylated. The sequenced Sobo element contains a single CCGG site but does not contain a CC(A/T)GG site. Southern hybridization detected signals only in the high molecular weight band in *Eco*RII- and *Hpa*II-digested DNA. However, several smaller bands were detected in *Bst*NI- and *Msp*I-digested DNA (Figure 6). These results indicate that the internal C of the CC(A/T)GG and CCGG sites are methylated.

DISCUSSION

Several lines of evidence suggest that Sobo is a recently amplified satellite repeat:

1. A survey of various Solanum species revealed that the Sobo element is present only in *S. bulbocastanum*. Several of these species, such as *S. cardiophyllum*, are closely related to *S. bulbocastanum* (SPOONER *et al.* 2004). In addition, the Sobo repeat is either absent or in a hemizygous condition in several *S. bulbocastanum* accessions (Table 1). These findings indicate that the Sobo repeat likely emerged sometime after the speciation of *S. bulbocastanum*.
2. Sequencing analysis of multiple subclones showed that the Sobo monomers are highly homogenized. We found only two subgroups of Sobo monomers that are distinguished by few single nucleotide polymorphisms. *Hin*dIII digestion of BAC 28G12, which contains an ~150-kb insert, resulted in only two bands (Figure 1), providing additional evidence of the homogeneity of the Sobo repeats. DNA sequences with no function are expected to accumulate mutations in the absence of selection (CHARLESWORTH *et al.* 1994). Therefore, lack of sequence divergence among different Sobo monomers may be explained by its recent amplification.
3. The Sobo monomer includes regions of similarity to several unrelated repetitive sequences (Figure 3). Thus, considering its sequence homogeneity, the Sobo monomer seems to originate by amplification of a single genomic locus containing these sequences.

Several mechanisms have been proposed to explain the origin and evolution of tandemly repeated DNA sequences. Unequal exchange between tandem arrays on sister chromatids or between arrays on homologous chromosomes has generally been assumed to be the most common mechanism of tandem repeat expansion or contraction (SMITH 1976). Unequal exchange, however, does not appear to explain the origin of the Sobo repeat. In general, recombination is rare in pericentromeric regions in Solanaceae species (SHERMAN and STACK 1995). Additionally, the Sobo repeat has only one FISH focus in several *S. bulbocastanum* accessions.

TABLE 1

Distribution of the Sobo repeat among *S. bulbocastanum* accessions

	Subspecies	Accession	Origin	Southern blot	Materials for FISH	Interphase FISH[a]
1	*bulbocastanum*	PI 243345C	Federal District, Mexico	ND	Tuber	No signal
2	*bulbocastanum*	PI 243505C	Federal District, Mexico	ND	Tuber	No signal
3	*bulbocastanum*	PI 243513B	Morelos, Mexico	ND	Tuber	No signal
4	*dolichophyllum*	PI 253210F	Morelos, Mexico	ND	Tuber	One signal
5	*bulbocastanum*	PI 275187A	Michoacan, Mexico	ND	Tuber	No signal
6	*bulbocastanum*	PI 275188E	Mexico	ND	Tuber	Two signals
7	*dolichophyllum*	PI 255516	Jalisco, Mexico	ND	Tuber	Two signals
8	*bulbocastanum*	PI 275185	Federal District, Mexico	Present	Seeds	Zero, one, two signals
9	*bulbocastanum*	PI 275192	Tlaxcala, Mexico	Present	Seeds	Zero, one, two signals
10	*bulbocastanum*	PI 275196	Oaxaca, Mexico	Present	Seeds	Two signals
11	*bulbocastanum*	PI 347757	Michoacan, Mexico	Absent	Seeds	No signal
12	*bulbocastanum*	PI 498223	Oaxaca, Mexico	Present	Seeds	Two signals
13	*partitum*	PI 275200	Huehuetenango, Guatemala	Absent	Seeds	No signal
14	*bulbocastanum*	PI 243510 (clone PT29)	Mexico Federal District	Present	Tuber	One signal

ND, no data.

[a] For lines that were obtained as seeds, we germinated 10 seeds from each line. Root tips were collected and mixed for cytological preparation. If nuclei with zero, one, or two FISH signals were all observed, it indicates that the original plant contains a hemizygous Sobo locus. If nuclei with either zero or two FISH signals were observed, it indicates that the original plant contains either no or a homozygous Sobo locus.

Our FISH technique should allow us to detect 5- to 10-kb (one to two Sobo monomer copies) DNA targets; thus there should be no technical limitations in identifying Sobo foci with the FISH approach. Therefore, our data indicate that in these accessions Sobo is in a hemizygous condition with no paralogous loci; there are no homologous or paralogous loci with which to recombine the Sobo repeat.

Extrachromosomal circular DNAs (eccDNA) have been discovered in a number of species (Gaubatz 1990; Cohen *et al.* 2003). Novel satellite repeats might originate from amplification of eccDNA through rolling circle replication, followed by reinsertion into the genome (Walsh 1987; Charlesworth *et al.* 1994). The origin of the Sobo repeat may be explained by this mechanism. The Sobo locus spans ~360 kb of highly homogenized DNA. In addition, the monomer size, 4.7 kb, is very large compared to most previously reported satellite repeats (Macas *et al.* 2002). Emergence of such a locus cannot be explained by other mechanisms, such as replication slippage, which would involve a short stretch of DNA in each cycle. The species-specific and hemizygous status of the Sobo repeat in several *S. bulbocastanum* accessions suggests that it is likely derived from a single amplification event, rather than from a series of events. If a new repeat is derived from mechanisms related to unequal exchanges or concerted evolution (Smith 1976; Dover 1986), it may require multiple events to eventually become a long and highly homogenized array.

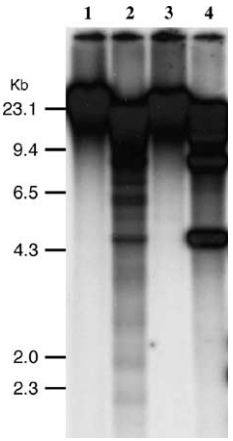


Figure 6.—Methylation analysis of the Sobo element. Genomic DNA isolated from *S. bulbocastanum* (PT29) was digested by *Eco*RII (lane 1), *Bst*NI (lane 2), *Hpa*II (lane 3), and *Msp*I (lane 4) and probed with the Sobo repeat.

We demonstrated that the major part of the Sobo monomer is related to the LTR of a *Sore1* element (Figure 3A). The structure of the Sobo repeat suggests that the original monomer may have originated by recombination-based excision of a genomic region between two LTRs separated with other repetitive sequences, giving rise to the eccDNA that could be amplified and reintegrated into the genome in a form of the long array of homogenous monomers. Surprisingly, the Sobo repeat does not hybridize to the rest of the *S. bulbocastanum* genome in both FISH and Southern hybridization. It suggests that the original Sobo monomer was derived from the LTR of an ancient and significantly diverged *Sore1* element. This hypothesis is supported by the fact that the *S. demissum Sore1* element, which shows se-

quence similarity with Sobo, is also an ancient element. Alternatively, potato genomes may contain very few copies of the *Sore1* retrotransposon.

Several previous reports demonstrate sequence similarities between satellite repeats and transposons (Pasero *et al.* 1993; Rossi *et al.* 1993; Langdon *et al.* 2000; Cheng and Murata 2003). These results suggest that transposons can be the original sources of satellite DNA. The supernumerary B chromosome of rye contains the E3900 satellite family (Blunden *et al.* 1993). The 3984-bp monomer of this satellite repeat contains an ~530-bp fragment that shares extensive similarity to the coding region for the *gag* protein of the *cereal* retrotransposon (Langdon *et al.* 2000). The remainder of the E3900 repeat appears to be unrelated to retrotransposon sequences. The monomer of another satellite repeat associated with the rye B chromosome, the D1100 family (Sandery *et al.* 1990; Houben *et al.* 1996), contains sequences related to Tnr1, a miniature inverted-repeat transposoable element reported in rice (Langdon *et al.* 2000). The sequence complexity of genomic clones containing the E3900 and D1100 repeats indicated that these two satellite families have undergone extensive amplification and rearrangements (Langdon *et al.* 2000). Therefore, it is not known if the transposon-related sequences were involved in the original formation of the satellite repeats or were later recruited in the E3900 and D1100 families. Cheng and Murata (2003) reported a centromere-specific tandem repeat in wheat that is likely derived from the centromere-specific retrotransposons. On the basis of the colocalization of the tandem repeat and the retrotransposons, most of the members of this tandem repeat are possibly still associated with the retrotransposons. It is not clear if this centromeric tandem repeat is organized as long arrays and whether these arrays have evolved independently from the centromeric retrotransposons.

In summary, although several previous reports showed sequence similarities between satellite repeats and transposons, it is not known if the transposon sequences contributed to the formation of the original monomers of the related satellite repeats. The Sobo repeat provides the first evidence that a satellite repeat can be amplified from a part of a retrotransposon in a sudden and dramatic fashion.

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